EMERGENCY MECHANICAL DECOKE

HF Sinclair Navajo — H-19 & H-20 Emergency Decoke

Job Report | USADeBusk | HFSinclair Navajo Refinery — Artesia, NM

JOB NO.	FACILITY	EXECUTION	PROJECT MANAGER
USA26038	HF Sinclair Navajo	Jul 10 – 17, 2026	Dacorey Slater
PO NO.	**SCOPE**	**HEATERS**	**DURATION**
PO-4200010828	Emergency Decoke	2	8 days

2	207	298	1
HEATERS CLEANED	OPERATING HOURS	PIGS RUN	SMART-PIG INSPECTION

Job Summary

Customer Details

FACILITY	HF Sinclair Navajo Refinery
ADDRESS	501 East Main St, Artesia, NM 88210
JOB & PO #	USA26038 / PO-4200010828
CONTACT	Travis Wilson

Project Details

SCOPE	Emergency mechanical decoke with filtration — H-19 Crude Charge & H-20 Vertical Crude Heaters
EXECUTION	July 10 – 17, 2026 (8 days)
HEATERS	H-19 Crude Charge · H-20 Vertical Crude Heaters
EQUIPMENT	(2) Trimax Pumpers (Trimax 3 & 5) · (2) Filtration Units · (2) Support Units · (2) Crew Trucks

Crew Details

PROJECT MANAGER	Dacorey Slater
SHIFT LEAD	Day: Jesse Utsey · Dacorey Slater | Night: Sam Mixon · James McDaniel
DAYSHIFT	Jesse Utsey, Dacorey Slater, Andre M., Blake B., Dandrew
NIGHTSHIFT	Sam Mixon, James McDaniel, Kailon, Daniel, Joel, Jordan

Project Duration

SCOPE	UNIT	RIG	PIG	SMART PIG	TOTAL
H-19 Crude Charge [1]	Trimax 5	9	74	6	89
H-20 Vertical Crude Hydro [2]	Trimax 3	20	98	0	118
TOTAL		**29**	**172**	**6**	**207**

[1] Delayed release — H-19 stood by until July 13 before rig-in. [2] As-built reconfigured to 4 coils looped to 2 passes; Trimax 5 rigged over to finish H-20 after H-19 completed.

Stand-By Summary

Stand-by is on-site time during which the pumping units were not actively pigging. Cause is recorded from the daily service receipts.

CAUSE	DATES	HOURS
Awaiting heater release, prep & crew call — HF Sinclair	Jul 11 – 12	58.0
Weather delay	Jul 12	2.5
Crossover TI removal & in-progress holds	Jul 13 – 15	10.0
Unit standing by — final rig-over	Jul 17	3.5
TOTAL		**74.0**

Combined project total: 207 operating hours plus 74 stand-by hours — 281 pumping-unit hours.

Pigs Used

298 pigs, 2.0" through 6.5". TC / pin pig. HR — Hell Razor. Foam — pin & gauge foam. Swab — dewatering swab. HC — honeycomb gauge.

SIZE	TC	HR	FOAM	SWAB/HC	TOTAL
2.0"	7				7
2.25"	12	1			13
2.3"	2				2
2.375"	6				6
2.5"	4				4
2.6"	1				1
2.625"	2				2
2.7"	1				1
2.75"	3	1			4
2.875"	2	3			5
3.0"	13	5	6		24
3.125"	11	4			15
3.25"	19	18			37

SIZE	TC	HR	FOAM	SWAB/HC	TOTAL
3.375"	1				1
3.5"		1			1
3.625"	3				3
3.75"	6	1			7
3.8"	1				1
3.875"	3	2	7		12
4.0"	7	1	5	2	15
4.125"	17				17
4.25"	27	22			49
4.5"		4			4
6.5"				32	32
Honeycomb				35	35

Heater Data and Results

H-19 — Crude Charge Heater

Number of Passes	2 coils looped to 1 pass (Pass 1 & 2)
Total Footage	~4,934' total (looped pig path ~5,754')
Convection Tube ID	4.026" (4.5" OD × 0.237, Sch 40)
Radiant Tube ID	3.068" (3.5" × 0.216) + 4.026" outlet section
Metallurgy	9 Chrome
Return Bends	180° U-Bends
Inlet / Outlet	(2) 4" 300# launchers at control valve station (grade)
Smart Pigging	Yes — Quest

Result: Moderately coked. The most severe deposits were in the 4" convection tubes, with the 3" radiant lightly fouled near the radiant inlet tubes. Passes 1 & 2 were pigged bi-directionally to buy-off.

> Dewatering the looped passes proved difficult; a 1600 CFM compressor failed to dewater the pass due to back pressure / tripped sensor. The temporary radiant-outlet loop was removed from the top of the can and each coil dewatered individually. Dewatering Pass 1 & 2 looped configuration will always fail regardless of the compressor; too much footage.

H-20 — Vertical Crude Hydro Heater

Number of Passes	As-built: 4 coils looped to 2 passes (temp 180's: Pass 1&3, Pass 2&4)
Total Footage	Convection 300' / pass · Radiant 1,134' / pass
Convection Tube ID	4.026" (4.5" × 0.237)
Radiant Tube ID	3.068" (3.5" × 0.216)
Metallurgy	9 Chrome
Return Bends	180° U-Bends
Inlet / Outlet	Convection (4) 4" launchers, 10' from grade · Radiant (2) 4" at grade
Smart Pigging	No

Result: Pass 4 was plugged prior to the decoke — its radiant coil was isolated and cleaned first, then the heater was re-looped to the 2-pass configuration for the remainder of the job. Fouling severity ran Pass 3 ≫ Pass 4 > Pass 1 ≈ Pass 2.

> Pass 3's radiant section was severely restricted with three separate heavy restrictions. Trimax 3 forced it open using its own pumps; 2" pigs run to establish flow returned badly damaged, and pigs were forced through one-way until enough flow was established to pig bi-directionally.

Flow Tests

H-19 — Pass 1 & 2 (Before | After)

GPM	BEFORE		AFTER		Δ PSI
	RPM	PSI	RPM	PSI	
400	1900	500	1770	420	80
350	1670	390	1530	320	70
300	1480	280	1350	250	30
250	1220	205	1100	165	40

H-20 — Pass 1 & 3 (Before | After)

GPM	BEFORE		AFTER		Δ PSI
	RPM	PSI	RPM	PSI	
250	1750	380	1220	225	155
200	1400	260	1150	160	100
150	1100	145	980	95	50
100	800	70	755	30	40

H-20 — Pass 2 & 4 (Before | After)

GPM	BEFORE		AFTER		Δ PSI
	RPM	PSI	RPM	PSI	
400	1700	388	1545	318	70
350	1506	305	1385	250	55
300	1298	229	1205	187	42
250	1084	169	1012	139	30

H-20 — Pass 2 & 4 Convection Only (Before | After)

GPM	BEFORE		AFTER		Δ PSI
	RPM	PSI	RPM	PSI	
550	1900	440	1770	370	70
500	1700	365	1585	305	60
450	1500	295	1400	245	50
400	1400	235	1315	195	40

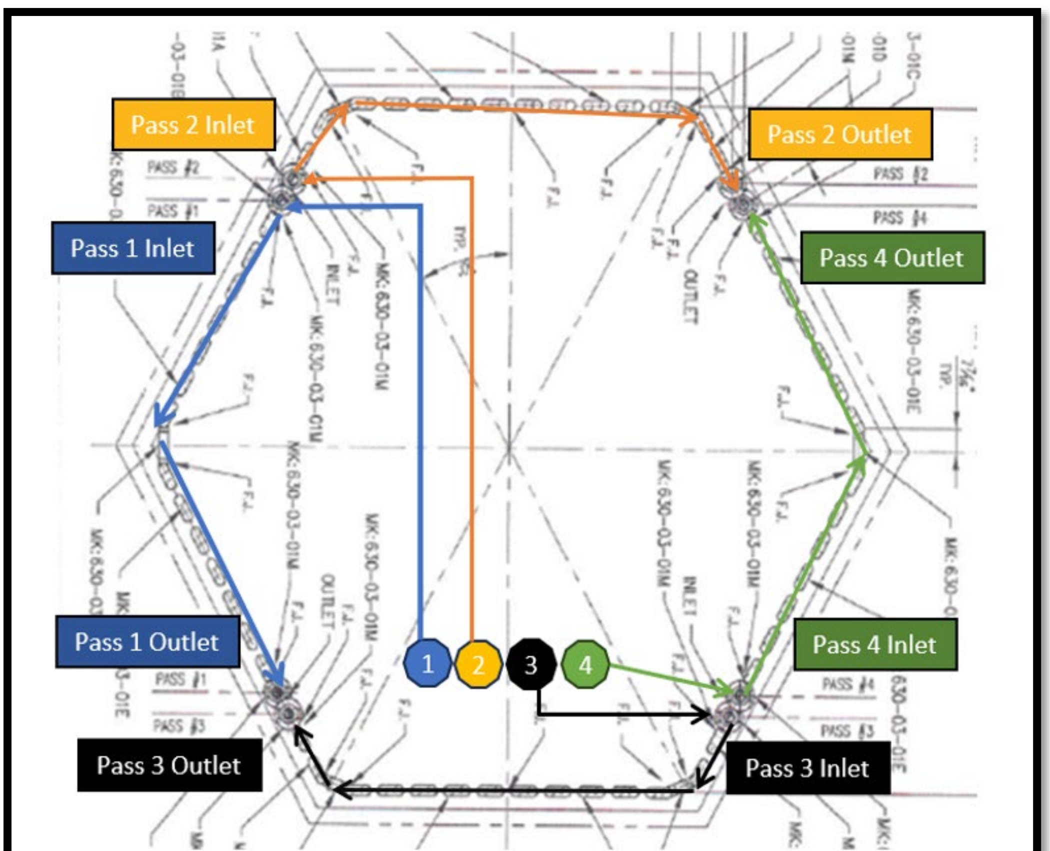
H-20 — Pass 4 Radiant Only (isolated) (Before | After)

GPM	BEFORE		AFTER		Δ PSI
	RPM	PSI	RPM	PSI	
400	1570	385	1280	237	148
350	1390	318	1132	193	125
300	1178	238	991	154	84
250	978	172	820	114	58

Images

H-20 Pass 3 — Pig Progression

H-20 Convection to Radiant Pass Visualization



Project Summary

USADebusk performed an emergency mechanical decoke of two fired heaters; H-19 Crude Charge and H-20 Vertical Crude, for HF Sinclair at the Navajo Refinery, July 10–17, 2026. Two Trimax pumping units worked in parallel, Trimax 5 on H-19 and Trimax 3 on H-20, each supported by its own filter press.

The project carried significant stand-by (74 hours): H-20 was released to USADebusk first, while H-19 remained on stand-by awaiting heater preparation through July 13, further extended by weather.

During the field-walk of H-20, we identified a plugged tube in the Pass 4 radiant section, a condition not confirmed prior to mobilization. The affected pass was isolated and cleared, and the heater's pass configuration was adapted in the field to clean the remaining sections efficiently. On H-19, a Quest smart pig confirmed only minimal residual coke following the mechanical clean.

Project Close

Both heaters were cleaned, inspected, and returned to HFSinclair. All equipment was decontaminated and demobilized from the facility on July 17, 2026.

USADebusk thanks HFSinclair Navajo Refinery team for their coordination and support throughout this emergency mobilization. We appreciate the opportunity to support your operation and look forward to continuing to serve your decoking and pigging needs.

Dacorey Slater

USADebusk
Project Manager
Cell: (832) 997-2072
dslater@usadebusk.com